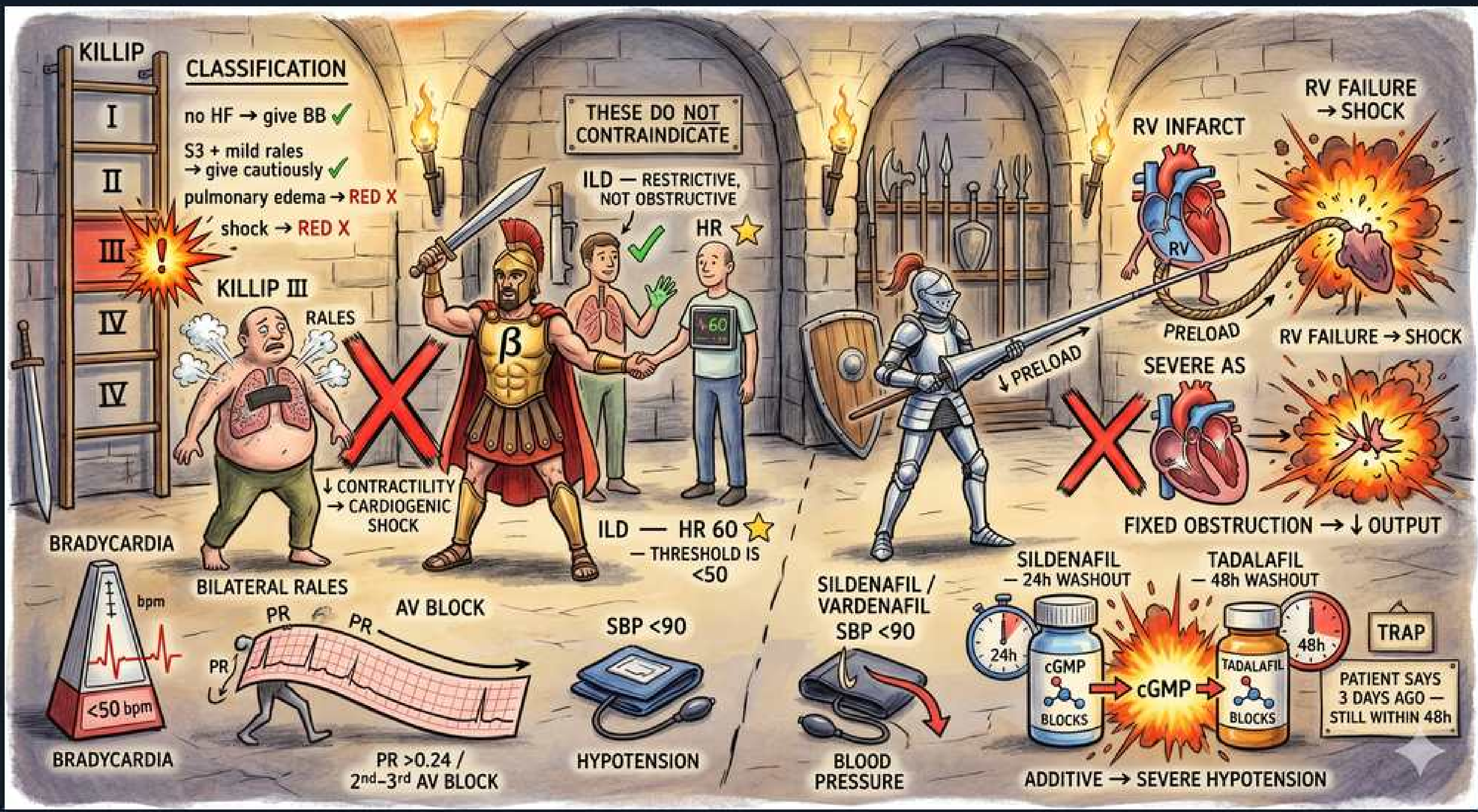
ACS — Killip Class, Beta-Blocker Cautions & Nitrate Traps



1. Killip classification

WHAT YOU SEE

Ladder 'KILLIP I–IV' with 'no HF → give BB □'

WHAT IT ENCODES

Killip class grades acute MI heart failure: I = no HF; II = rales/S3; III = pulmonary edema; IV = cardiogenic shock. Higher class = higher mortality and modifies therapy (BB timing).

BOARD TRAP

Memorize the gradient: Killip IV (cardiogenic shock) has the highest mortality and is a contraindication to early IV beta-blockade.

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2. Killip I — give beta-blocker

WHAT YOU SEE

'no HF □ give BB'

WHAT IT ENCODES

In Killip I (no failure, hemodynamically stable) start an oral beta-blocker early — it reduces ischemia, reinfarction, and arrhythmic death post-MI.

BOARD TRAP

Beta-blockers help only the stable patient; giving them to a decompensated MI causes harm.

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3. Killip II — give cautiously

WHAT YOU SEE

'S3 + mild rales — give cautiously / pulmonary edema RED X / shock RED X'

WHAT IT ENCODES

With mild failure (S3, basal rales) introduce beta-blockers cautiously and low-dose. Overt pulmonary edema (Killip III) and shock (Killip IV) are contraindications to early beta-blockade.

BOARD TRAP

Pushing a beta-blocker into early pulmonary edema or shock — COMMIT trial showed early IV BB increases cardiogenic shock.

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4. Killip III — pulmonary edema (no early BB)

WHAT YOU SEE

'KILLIP III' rales figure, RED X

WHAT IT ENCODES

Killip III (frank pulmonary edema) — defer beta-blockers; treat HF first. Early aggressive beta-blockade in acute decompensated MI precipitates cardiogenic shock.

BOARD TRAP

Treating Killip III edema as a green light for rate control — it is a contraindication to early IV beta-blocker.

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5. Bradycardia / HR <50 — no BB

WHAT YOU SEE

Metronome '<50 bpm BRADYCARDIA'

WHAT IT ENCODES

Bradycardia (HR <60, threshold ~<50) is a contraindication to beta-blockers — they further slow the rate and can precipitate complete block, especially with inferior MI/RCA AV-node involvement.

BOARD TRAP

Beta-blocking an inferior MI with bradycardia or high-grade AV block — risk of hemodynamic collapse.

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6. AV block — PR >0.24 / 2nd-3rd degree

WHAT YOU SEE

'AV BLOCK — PR >0.24 / 2nd–3rd AV BLOCK'

WHAT IT ENCODES

PR >0.24 s (1st-degree), or 2nd/3rd-degree AV block contraindicate beta-blockers. Inferior MI (RCA) commonly causes AV-nodal block — usually transient and atropine-responsive.

BOARD TRAP

Adding a beta-blocker on top of 2nd/3rd-degree AV block — can progress to complete heart block and asystole.

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7. SBP <90 / hypotension — no BB

WHAT YOU SEE

'SBP <90 HYPOTENSION'

WHAT IT ENCODES

Systolic BP <90 mmHg (hypotension) is a beta-blocker contraindication — negative inotropy worsens output and can tip the patient into shock.

BOARD TRAP

Giving a beta-blocker to a hypotensive MI patient — wait until hemodynamically stable.

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8. These do NOT contraindicate

WHAT YOU SEE

Banner 'THESE DO NOT CONTRAINDICATE' with checkmarks

WHAT IT ENCODES

Reactive airway disease can usually tolerate cardioselective beta-1 blockers; well-controlled HR and stable HFrEF on therapy are not absolute contraindications. Don't withhold prognostic beta-blockade unnecessarily.

BOARD TRAP

Withholding a beta-blocker for mild stable COPD/asthma — cardioselective agents (metoprolol, bisoprolol) are usually safe.

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9. ILD restrictive, not obstructive

WHAT YOU SEE

'ILD = RESTRICTIVE, NOT OBSTRUCTIVE'

WHAT IT ENCODES

Restrictive lung disease (ILD) has no bronchospastic component — beta-blockers are fine. The caution is bronchospasm in obstructive disease, and even there cardioselective agents are tolerated.

BOARD TRAP

Confusing restrictive ILD with obstructive asthma — ILD is not a reason to withhold beta-blockade.

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10. HR 60 threshold <50

WHAT YOU SEE

'HR 60 — THRESHOLD <50'

WHAT IT ENCODES

A resting HR ~60 is acceptable to start/continue a beta-blocker; the hard stop is HR <50 (symptomatic bradycardia). Titrate to rate and symptoms.

BOARD TRAP

Stopping a beta-blocker at HR 60 — the contraindication threshold is symptomatic bradycardia <50.

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11. RV infarct — preload dependent

WHAT YOU SEE

'RV INFARCT → SHOCK … PRELOAD' heart + explosion

WHAT IT ENCODES

RV infarction (inferior STEMI, RCA) is preload-dependent — give IV fluids, AVOID nitrates/diuretics/morphine which drop preload and cause profound hypotension. Treat bradycardia, support RV.

BOARD TRAP

Giving nitroglycerin to an inferior MI with RV involvement — preload drop → severe hypotension/collapse.

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12. Severe AS — fixed obstruction

WHAT YOU SEE

'SEVERE AS — FIXED OBSTRUCTION → ↓OUTPUT'

WHAT IT ENCODES

Severe aortic stenosis is a fixed obstruction; nitrate-induced preload/afterload drop can crash cardiac output. Use vasodilators very cautiously — same preload-sensitive caution as RV infarct.

BOARD TRAP

Routine nitrates in severe AS chest pain — venodilation drops preload across a fixed valve and causes syncope/hypotension.

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13. Sildenafil/vardenafil 24h washout

WHAT YOU SEE

'SILDENAFIL/VARDENAFIL SBP <90 — 24h washout; cGMP'

WHAT IT ENCODES

PDE-5 inhibitors potentiate nitrate-induced cGMP → catastrophic hypotension. Sildenafil/vardenafil need a 24-hour nitrate-free window; tadalafil (long-acting) needs 48 hours.

BOARD TRAP

Giving nitrates within 24h of sildenafil (48h for tadalafil) — additive cGMP → severe, refractory hypotension.

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14. Tadalafil 48h washout

WHAT YOU SEE

'TADALAFIL 48h WASHOUT'

WHAT IT ENCODES

Tadalafil's long half-life mandates a 48-hour wait before nitrates. The classic trap is the patient who took it 'days ago' but is still within window.

BOARD TRAP

'Took it 3 days ago, still within 48h' — verify timing; tadalafil washout is 48 hours, not 24.

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15. Nitrate + PDE5 = additive hypotension

WHAT YOU SEE

'ADDITIVE → SEVERE HYPOTENSION' explosion

WHAT IT ENCODES

Nitrates (NO donor) and PDE-5 inhibitors both raise cGMP — combination causes profound, refractory hypotension. Treat with fluids and alpha-agonists if it occurs; prevent by taking a careful drug history.

BOARD TRAP

The exam vignette where chest-pain nitrate is given to a man on a PDE-5 inhibitor — the answer is severe hypotension.
